



FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

TER6313 Enterprise Resource Planning

Trimester October/November 2024

Report

Dr. Siti Fatimah binti Abdul Razak

Group Leader:

Student ID	Student Name	Programme	Lab Section
1211203835	Liew Zhi Tong	BIA	1B

Group Members:

No.	Student ID	Student Name	Programme	Lab Section
1	1211201569	Yeo Hui Yi	BIA	1B
2	1191202283	Uithiswary a/p Velan	BIA	1C
3	1211204549	Priveeta Gobi Krishnan	BIA	1B

Declaration

WE hereby declare that all group members' names are correctly included in the above section and all members agreed to equal marks attainment. We hold a copy of this assignment which we can produce if the original is lost or damaged. We certify that not part of this final assessment has been copied from any other student's work or from any other source except where due acknowledgement is made in the report.

WE also acknowledge the use of AI tools (if any) is aimed at brainstorming, rephrasing and enhancing decision-making processes. We affirm our commitment to ethical guidelines, transparency, and responsible utilization of AI technology in acquiring solutions.

Table of Contents

Role of Members.....	3
1.0 Introduction to Business	4
2.0 Current Business Processes.....	5
3.0 Challenges & Inefficiencies	7
4.0 Specific SAP ERP Modules Relevant to the Observations.....	8
4.1 Mapping SAP SD Module to Identified Challenges.....	9
4.2 Justification for Selecting the SAP SD Module	10
4.3 Key ERP Technologies for SAP SD Implementation	11
4.4.1 SAP S/4HANA – The Digital Core	11
4.4.2 SAP Fiori – User-Friendly Interface	12
4.4.3 SAP Business Technology Platform (BTP) – Seamless Integration & Automation ...	12
4.4.4 Cloud Platforms (AWS, Azure, Google Cloud) – Scalable & Secure Infrastructure .	12
4.5 Summary.....	12
5.0 ERP Implementation Plan	13
6.0 Expected Benefits & SDG Impact.....	15
7.0 Conclusion	19
References	21
Appendix.....	21
The clear image for BPMN business process diagram	21

Role of Members

No	Student ID	Student Name	Task Contribution
1.	1211203835	Liew Zhi Tong	- Assigned tasks to team members, ensured report format and content were aligned with project requirements. - Led discussions on food court observations, business processes, SDG ERP solution selection, and ERP implementation steps. - Completed the Introduction and Specific SAP ERP Modules Relevant to the Observations report sections.
2.	1211201569	Yeo Hui Yi	- Participated in discussions on food court observations, business processes, SDG ERP solution selection, and ERP implementation steps. - Completed the Explanation of business process and ERP Implementation Plan.
3.	1191202283	Uithiswary a/p Velan	- Participated in discussions on food court observations, business processes, SDG ERP solution selection, and ERP implementation steps. - Completed the BPMN diagram of business process and Conclusion.
4.	1211204549	Priveeta Gobi Krishnan	- Participated in discussions on food court observations, business processes, SDG ERP solution selection, and ERP implementation steps. - Completed the Challenges & Inefficiencies and Expected Benefits & SDG Impact.

1.0 Introduction to Business

Madam Yan Foodcourt has been in operation for over 1.5 years, serving the public with rented-out stalls. Located in a strategic position the food court serves three distinct customer groups including office workers and university students and families, the food court operates from 11 AM until 12 AM to provide an active dining setting. It offers an efficient drink-ordering system in combination with well-ventilated air and pleasant staff which supports dining and social activities. However, like many conventional food courts, Madam Yan Foodcourt faces operational challenges, including a shortage of seating during peak hours, inconsistent food quality, and a need for improved cleaning services.

This report intends to elaborate on how the ERP system, specifically SAP ERP, can transform operations at Madam Yan Foodcourt by addressing these challenges and inefficiencies. ERP systems, by streamlining processes, enhancing customer experience, and ensuring proper resource allocation, provide a comprehensive solution for optimizing the food court's operations.

During visits and observation, we found that I.T can be effective in many areas by putting an ERP system in place from effective communication to better inventory management and even staff coordination. In this report, we will outline the assessment of the current business processes, their challenges and inefficiencies, along with a proposal including the customization of SAP ERP modules to suit the food court's particular needs. Moreover, the report provides a roadmap for implementation and highlights the benefits, focusing specifically on sustainability and social attribution in the framework of the United Nations' Sustainable Development Goals (SDGs).

This case study demonstrates how, with a focus on innovation and efficiency, leveraging SAP ERP can elevate Madam Yan Foodcourt's operational excellence and set a benchmark for other food courts.

2.0 Current Business Processes

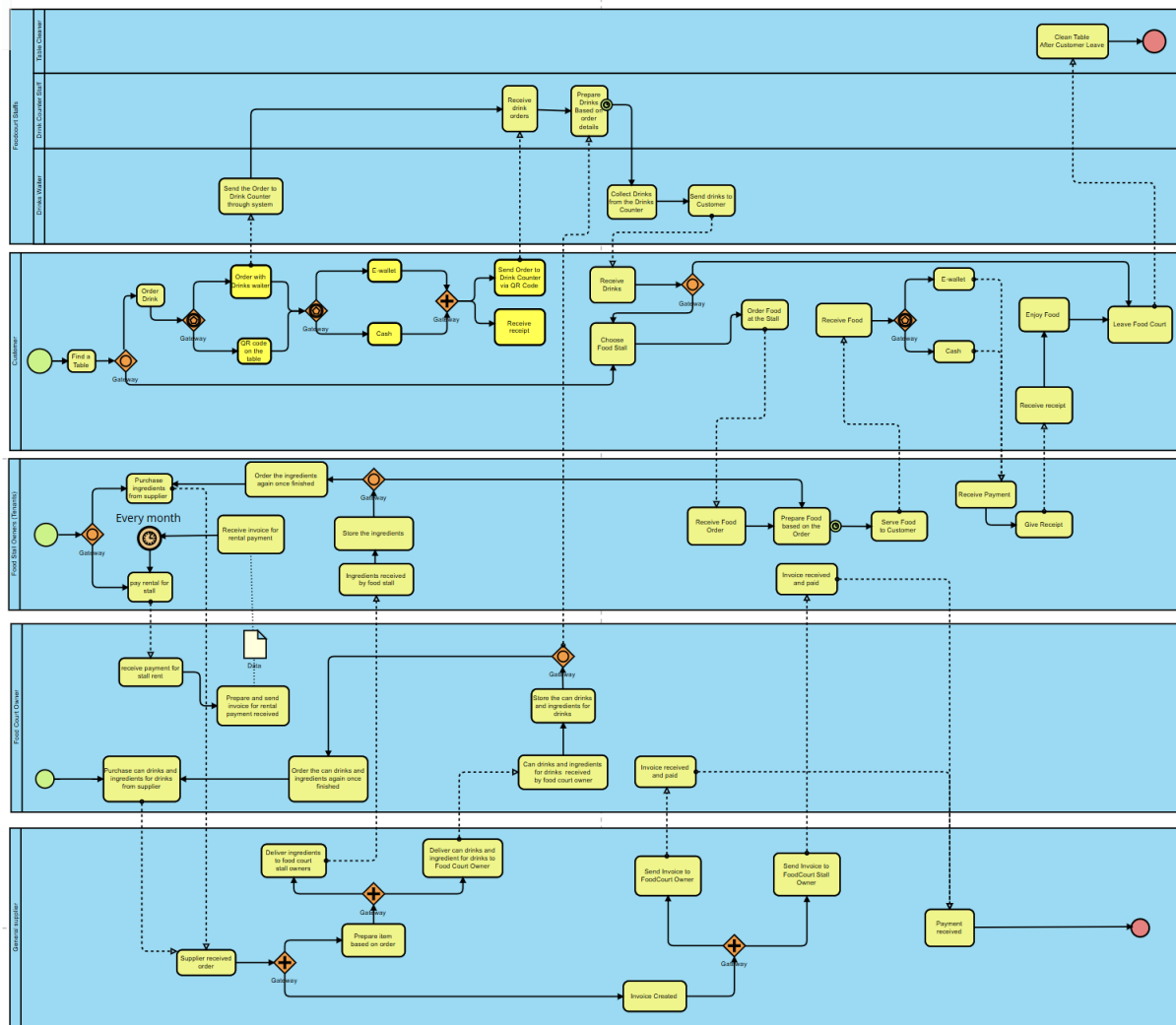


Figure 1: The business process is illustrated using a BPMN diagram

The following section explains the business process based on Figure 1 BPMN diagram:

Customer ordering and payment process

- Customer finds a table, either order drink or choose food stall to order food.
- If customers order drinks, they can either place their order through the drinks waiter, who will enter it into the system, or scan the QR code on the table to order directly.
- After drinks ordering, payment made using either E-wallet or cash.
- After payment, the order will be sent to the drink counter, and the customer will receive a receipt.
- If customer orders food, they can choose food stall, order food at the stall and order will be received by the food stall owners.

- After customer received food, they can choose either pay by E-wallet or cash and receive receipt.
- Customer enjoys their food and leaves the food court. Table cleaner will clean the table after customer leaves the food court.

Food and beverages serving process

- The food stall owners receive food order, prepare order and serve food to customer.
- After receiving food payment, provide receipt to the customer.
- After drinks ordering with waiter, order will be sent to drink counter through system.
- Drink counter receives and prepare drinks order. Then, waiter collects drinks at the counter and send to customer.

Ingredients purchased process

- The food stall owners purchase ingredients from supplier every day. Once finished, order will be made to supplier.
- The food court owner purchase canned drinks and ingredients from supplier. Once finished, order will be made to supplier.

Ingredients supplied process

- General supplier received order from food stall owners and food court owner and prepare order. Then, deliver ingredients to food court stall owners and canned drinks and ingredients to food court owner.
- After order received, invoice will be created and sent to food court owner and food stall owners. Then, payment will be made and received.
- Ingredients will be received by food stall owners and they store it.
- Canned drinks and ingredients will be received by food court owner and they store it.

Stall rental process

- The food stall owners pay a rental fee to food court owner, then food court owner receive rental fee.
- The food court owner prepares and sends an invoice for the rental payment received, and the food stall owners receive the invoice.
- This process will be generated every month.

3.0 Challenges & Inefficiencies

In accordance with the business process shows several obstacles and inefficiencies that have an impact on the overall operation and clientele's experience. These issues, which lead to operational delays and irregularities in service delivery, are caused by a confluence of disjointed workflows, a dependence on manual interventions, and a lack of technological integration.

The dual method of ordering drinks is a major inefficiency. The alternate pathway, where consumers who are unable to utilise the QR code must rely on the drinks waiter, adds variability to the process even if the QR code system offers a simplified and technologically advanced means for customers to place orders directly. Since the waiter's assistance is only needed if the primary system malfunctions, this dependency on manual intervention for some customers leads to an imbalance in the workflow. When waiters must handle several requests at once, especially during peak hours, this causes delays and disrupts the otherwise automated process. This approach adds an unpredictable layer to the operation, even if it is required to accommodate all consumers.

Another crucial issue is the division of the ordering procedures for food and beverages. Customers are needed to manage these interactions on their own, placing drink orders via a drinks waiter or the QR code system before proceeding to individual food stalls to place meal orders. Because of this fragmentation, patrons must go through two different procedures for what is ultimately a single dining experience. Because customers must do comparable tasks for both food and drinks—such as selecting payment methods and keeping track of receipts—a fragmented system like this leads to inefficiencies. There are gaps in the operation's efficiency and flow since there is no centralisation in tracking customer preferences, orders, or payments due to the absence of a single ordering system.

Potential inefficiencies are also brought about by the two payment methods—cash and e-wallet. Although providing both choices serves a wider clientele, complexity is increased by the lack of a completely integrated payment mechanism. E-wallet transactions rely on the customer's ability to scan QR codes and use mobile devices efficiently, whereas cash purchases are slower because they entail handling actual currency and giving change. Transaction timings become variable as a result, which could cause delays, especially during peak hours when lines

start to form. Payments for food and beverages are handled separately, which further shatters the client experience and raises the possibility of financial reconciliation problems.

Additionally, the food court's general efficiency in operation is decreased by the workflows for food and drink preparation not being coordinated. There is no centralised system to coordinate the timing of order fulfilment; each procedure runs independently. The eating experience may be disrupted because of patrons receiving their drinks either long before or after their food. Customers are irritated by this lack of integration, and employees become inefficient having to handle two different workflows instead of a single system.

Finally, the table preparation and cleaning procedure is isolated from the remainder of the workflow. The upkeep of cleanliness is the responsibility of the table cleaners, but there is no mechanism in place to guarantee that table turnover corresponds with consumer flow. Delays in cleaning tables during peak hours might make it difficult to seat new patrons on schedule, which further slows down business operations. The food court's ability to effectively manage high client volumes is hampered by this lack of coordination.

As a result of fragmented workflows, inconsistent technology utilisation, and a lack of integration, Madam Yan's food court business process declines. These inefficiencies have an impact on worker productivity as well as the customer experience. To solve these problems, a more integrated system that lessens the need for manual intervention while integrating ordering, payment, and seating management is needed. Enhancing client happiness, optimising resource utilisation, and streamlining operations are all benefits of such enhancements.

4.0 Specific SAP ERP Modules Relevant to the Observations

To address the challenges and inefficiencies identified in Madam Yan's food court operations, the SAP Sales and Distribution (SD) module is proposed as the most suitable ERP solution. This module offers a comprehensive system to streamline order processing, enhance payment integration, and improve overall workflow efficiency.

4.1 Mapping SAP SD Module to Identified Challenges

The SAP SD module is designed to handle key aspects of order management, sales processing, and customer transactions. Table 4.1.1 maps the SAP SD module functionalities to the operational challenges observed in the food court.

Table 4.1.1 Mapping SAP SD Module Features to Identified Challenges and Inefficiencies

Challenges & Inefficiencies	SAP SD Module Features	How It Solves the Issue
Dual method of ordering drinks (QR code vs. manual waiter system) causes delays and disrupts workflow balance.	Centralized Order Processing	The SAP SD module streamlines food and drink orders into a single digital system, minimizing the need for manual intervention. For those who pay through self-service kiosks or mobile apps, they get a perfect experience.
Separate ordering process for food and beverages, causing inefficiencies in tracking customer preferences and payments.	Integrated Sales Order Management	This module facilitates real-time order synchronization and allows food and beverage transactions to be managed under one roof. This enhances the convenience for customers while allowing staff to be more effective across functional areas.
Fragmented payment system (cash vs. e-wallet) causes delays and reconciliation issues.	Automated Payment Processing	SAP SD integrates multiple payment methods (cash, e-wallets, and credit/debit cards) within a single transaction platform, ensuring smooth and accurate payment handling.
Uncoordinated workflows for food and drink preparation, leading to	Delivery & Billing Integration	The system can automatically track order status so that drinks and meals can be prepared and delivered together

inconsistencies in order fulfilment.		in the same order. This enhances the overall dining experience.
Inefficient table turnover due to uncoordinated cleaning processes.	Customer Queue & Order Tracking	SAP SD allows managers to monitor real-time table occupancy and turnover, ensuring cleaning staff is promptly notified to prepare tables for incoming guests.

4.2 Justification for Selecting the SAP SD Module

The SAP Sales and Distribution (SD) module is the best solution for food court operational challenges as it can best manage sales orders, payments, and customer transactions. SAP SD simplifies business processes, minimizes redundancies, and ensures efficient and timely service delivery through an integrated, automated data system that encompasses all these key functions.

One of the primary advantages of the SAP SD module is its mainly known for providing end-to-end order management. It integrates food and beverage orders into one digital module, forgoing the choppy traditional ordering process. Unlike other self-ordering systems, the integration with General Prism directly reduces the amount of manual interaction needed by customers to place orders through kiosks or mobile apps. So, waitstaff have less work to do and will primarily focus on serving guests as order-taking will be hassle-free and seamless, thereby improving operational efficiency.

Apart from order management, SAP SD module optimizes these payments and financial reconciliation even further within an automated financial system that connects all transaction types, whether cash, e-wallet, and card payment. This reduces human error and accelerates the checkout process to enable more rapid transactions and ensure accurate tracking of revenue. The system also improves financial transparency for the food court by minimizing variances in financial statements, thus improving financial sustainability and reconciliation.

Additionally, the SAP SD module streamlines workflow efficiency and improves the customer experience by coordinating food and drink preparation automatically. Resh Mane (0)

Yes, orders are handled in sync, so the food and drinks are always prepared and discovered at the same time, so there are zero service discrepancies. Also, thanks to the real-time data tracking functionality, managers can keep track of customer orders to minimize waiting times and make data-driven decisions to improve service quality.

Implementing SAP SD also plays an important role in improving customer satisfaction and business growth. It is a seamless digital ordering and payment system which makes their convenience a lot higher who requires high customer retention rate. In addition, the centralized data system allows the businesses to monitor customer preferences and purchasing patterns, so they implement focused promotions and loyalty programs encouraging repeat visits. By harnessing this data, businesses can not only improve the customer experience but also contribute to sustained successful business performance.

In short, the SAP SD module tackles major inefficiencies related to order, payment, and workflow synchronization, and it certainly emerges as the best ERP system to opt for to suit the food court. This can serve as a viable solution to most problem statements of Food court as this solution can optimize the operations by minimizing the manual interventions, creating insights from data, thus optimizing the overall operations and contribute to better customer satisfaction and optimization of business.

4.3 Key ERP Technologies for SAP SD Implementation

To ensure the successful deployment of the SAP Sales and Distribution (SD) module, several key ERP technologies are essential. These technologies enhance system performance, improve user experience, and support seamless integration with cloud-based infrastructure. The core technologies that complement SAP SD include:

4.4.1 SAP S/4HANA – The Digital Core

SAP SD operates within SAP S/4HANA, an advanced ERP suite that provides:

- **High-speed data processing:** Ensures real-time tracking of food and beverage orders.
- **Advanced analytics:** Helps predict peak hours, allowing better workforce and inventory management.

- **Integrated order processing:** Enhances customer experience by reducing delays and ensuring smooth transactions.

4.4.2 SAP Fiori – User-Friendly Interface

SAP Fiori improves user interaction by offering:

- **Simplified UI:** Provides an intuitive experience for order management and transaction tracking.
- **Mobile accessibility:** Enables waitstaff and customers to interact with the system via mobile devices.
- **Customization:** Allows the food court to tailor dashboards for order monitoring and sales performance analysis.

4.4.3 SAP Business Technology Platform (BTP) – Seamless Integration & Automation

SAP BTP acts as the foundation for extending SAP SD functionalities:

- **Cloud and IoT Integration:** Supports cashless payment methods and real-time transaction updates.
- **Predictive analytics & AI:** Helps optimize sales strategies and customer engagement.
- **Workflow automation:** Reduces manual intervention by enabling automatic order routing and billing.

4.4.4 Cloud Platforms (AWS, Azure, Google Cloud) – Scalable & Secure Infrastructure

The implementation of SAP SD is further strengthened by cloud-based hosting solutions:

- **Scalability:** Easily accommodates growing transaction volumes without hardware constraints.
- **High availability & security:** Ensures system uptime and protects customer data.
- **Remote access:** Allows business owners to monitor operations from any location.

4.5 Summary

By implementing SAP Sales and Distribution (SD) module will allow Madam Yan's food court to optimize order processing as well as payment integration with enhanced workflow efficiency. The module operates to enhance business efficiency through operation streamlining

and developed financial tracking features along with improved customer experience which addresses previous operational limitations.

Furthermore, the key ERP technologies—SAP S/4HANA, SAP Fiori, SAP Business Technology Platform (BTP), and cloud platforms (AWS, Azure, Google Cloud)—form the backbone of this implementation. These technologies establish a digital system which provides powerful fast operation and scale and security for managing orders in real-time with automated payments and workflow control. The food court will enhance its operations and long-term business success through improved efficiency and customer satisfaction because it employs this integrated ERP solution.

5.0 ERP Implementation Plan

Phase 1: Planning & Requirement Analysis (Month 1 - 2)

1. Define objectives & scope

- Implement SAP Sales and Distribution (SD) module to facilitate the ordering and payment process and work cycle.
- Centralized order processing, automatic generation of payment and real-time tracking of orders are integrated into one system.

2. Stakeholder consultation

- Meetings with food stall owners, employees and suppliers to know their requirements.
- Identify problems related to the current manual process and set their expectations.

3. System requirements & customization needs

- Identify integration requirements with QR code ordering, E-wallet and food stall POS systems.
- Customize order synchronization between drinks counters and food stalls.

Phase 2: System design & configuration (Month 3 - 4)

1. ERP system design

- Design a centralized ordering system to integrate food and beverage transactions.
- Configure SAP SD to support multi-payment options like cash, credit card and E-wallet.

2. Data migration plan

- Migrate existing sales and financial data to SAP SD.
- Ensure order history and payment logs are correctly mapped.

3. User interface and access control

- Design a simple dashboard for food stall owners to monitor their orders and sales.
- Set up user roles such as food stall owners, drink counters and cleaners.

Phase 3: Testing and staff training (Month 5 - 6)

1. Testing at selected stalls

- Implement SAP SD in selected stalls to test the efficiency.
- Monitor system performance and identify technical issues.

2. Training programs

- Conduct training on the management of orders and reconciliation of their payment.
- Practical training in system navigation and troubleshooting.

3. Collect customer feedback

- Gather feedback on ordering experience from early users.
- Identify potential system improvements before full deployment.

Phase 4: Full implementation and optimization (Month 7 - 8)

1. Go-Live Across Stalls

- Deploy SAP SD across all food stalls and drink counters.
- Ensure smooth order synchronization between food and beverages.

2. Monitor system performance

- Identify technical issues and optimize order processing speed.
- Ensure real-time data tracking for payments and inventory.

3. Optimize workflows

- Automate table cleaning notifications after customer leaves.
- Reduce waiting time by synchronizing food and drink preparation.

Phase 5: Continuous improvement and maintenance (Month 9 - 12 & beyond)

1. Ongoing staff supports and updates

- Regular technical support and updates for the food stall owners.
- New feature introduction on operational requirements.

2. Monitoring and analytics performance

- Apply SAP SD analytics for peak hour information, order trends and sales data.
- Apply customer loyalty programs based on purchasing behaviour.

3. Increased integration and future enhancements

- Integrate AI-driven demand forecasting to do better inventory management.
- Explore additional modules for HR Staff Scheduling and Finance Cost Tracking.

6.0 Expected Benefits & SDG Impact

With the implementation of the SAP Sales and Distribution (SD) Module at Madam Yan Foodcourt a bright future for operational efficiency, customer experience and financial performance lies ahead. Through the solution of the current business process inefficiencies, the ERP will make the food court's operations more smooth, modern, streamlined, sustainable and pleasant.

Key expected benefits are the following:

- **Streamlined Order Processing**

To centralize food and beverage orders into a single digital system, the SAP SD module will be used. This will eliminate the current two ordering system (manual waiter vs. QR code) and save on manual intervention while speeding up the whole ordering process. This will consequently bring the order processing times down significantly thereby leading to faster and more accurate service, especially during peak hours.

- **Improved Payment Efficiency**

Multiple payment options such as cash, e-wallets, and credit/debit card will be integrated into one platform which makes payment process easier. This means the transactions will take less time, the errors in financial reconciliations will also be minimized, and customers will have less friction, a more convenient time making payments. As a result, payment processing will be automated, the financial transparency will be enhanced along with reducing the risk of revenue leakage.

- **Enhanced Customer Satisfaction**

Real time order tracking and synchronization of food and drink preparation will lead to customers getting their orders faster and with a higher level of consistency. This will

allow the system to coordinate meal and drink delivery so that customers will have the complete order without an unnecessary delay. It will result to higher customer satisfaction, more repeat visits and good positive word of mouth referrals.

- **Optimized Resource Utilization**

The SAP SD module would enhance food and drink preparation coordination, which would eliminate waste and reduce the use of resource. The system will allow the food court to optimise ingredient usage by minimising overstock, control demand and reduce food wastage, by presenting real time information of inventory levels and demand patterns. It will limit the costs and increase profitability.

- **Data-Driven Decision Making**

The centralized data system will give a lot of insight into the data that customers buy, preferences, and purchasing patterns as well as the operational data. The information in this data piece can be used to deliver targeted promotions, loyalty program and menu adjustments in reaction to preferences of customers. Through this data, Madam Yan Foodcourt can keep its customers and boost sales and better decide on important business performance aspects.

- **Improved Staff Productivity**

The SAP SD module will automate repeatable tasks like order entry, cash taking by doing so it will take time of the staff and make them engaged more to the things like customer service and food preparation. With this in place, staff productivity will be increased with resulting reduction of burden of work and increasing job satisfaction, which leads to a more motivated and efficient force.

The proposed ERP solution aligns with several United Nations Sustainable Development Goals (SDGs), emphasizing sustainability, social responsibility, and innovation.

- ❖ **Goal 3, Good Health and Well-Being - The goal is the ensure healthy lives and promote well-being for all customers.**

This is aimed at increasing the preparation and the delivery of food in the most efficient way so that customers can get them fast and good. By doing this, it helps to lower the risk of food borne illness and improves the dining experience for the customer as they dine out. Also, the system allows monitoring of use of ingredients and expiration dates, which oversees food safety standards such that one can only use fresh and safe ingredients in food preparation.

❖ **Goal 8, Decent Work and Economic Growth – To promote sustained, inclusive, and sustainable economic growth, full and productive employment**

The SAP SD module automates redundant tasks, which increases staff's productivity and the overall job satisfaction. Order processing will no longer require employees to spend time on this job, freeing up their time to interact with customers and prepare food. This helps to enhance the food court's economic growth as well as making it possible for the work that is carried out to be done decently within a decent work environment.

❖ **Goal 9, Industry, Innovation and Infrastructure - Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.**

This technological advancement with SAP SD represents a huge digital advantage that Madam Yan Foodcourt now has, replacing the outdated, manual processes with a more modern way of running their foodcourt. The system helps remove dependence on physical resources by removing paper-based orders and automating workflows of the food court. Such efforts help in sustainable industrialization and feature the food court as an innovative position in the industry.

❖ **Goal 10, Reduced Inequalities – To reduce inequality within and among countries**

This integration of varying payment methods (cash, e-wallets, and credit/debit cards) within the SAP SD module allows all customers to have a clean dine, with no matter what payment method they prefer. This inclusivity eliminates obstacles to customers that may not have the ability to employ digital payment methods making it more unbiased and useful. In addition, the system's unique capability to offer real time data on the preferences and patterns of preferences and purchasing of customers by the food court helps the food court provide diversely customer's ability of service delivery.

❖ **Goal 11, Sustainable Cities and Communities – To make cities and human settlements inclusive, safe, resilient, and sustainable**

Madam Yan Foodcourt makes a positive change in helping to create a more sustainable community by improving the efficiency and sustainability of its operations. The food court helps customers be served quick and efficiently, at minimal loss, contributes to making city everyone living condition better, and creates a standard of the living conditions for other businesses in city.

❖ **Goal 12, Responsible Consumption – Ensure sustainable consumption and production patterns**

This goal directly connects to SAP SD module's capacity to streamline inventory management and prevent food waste. The system provides real time data on the

ingredient usage and demand patterns, which enable the resources to be used more efficiently, something that can be achieved through reducing the over production and waste of it. It promotes the sustainable consumption and production, helping the food court reduce the excess wastage on the environment.

Key metrics for success of the SAP SD module implementation:

- **Order Processing time**

This is one crucial metric, at this point the average order processing time is 10 minutes, and the processing is often delayed during peak hours as order processing is manual, and workflows are fragmented. The accelerated order processing time is intended to drop by 30 percent—down to 7 minutes per order—by rolling out the SAP SD module’s centralized and automated order management system. The result of this improvement would be to speed up service, particularly on the busiest hours, increasing customer satisfaction and table turnover. In addition, it will eliminate customers’ wait time in a big way and make the whole dining experience much better.

- **Payment Efficiency**

Today, payment processing averages at 5 minutes per transaction and suffer delays because the payments system has fragmented and manual reconciliation processes. It is expected to integrate various payment methods (cash, e-wallets and credit and debit cards) on a single automated platform, reducing payment processing time to 40% with an achievement rate of 3 minutes per transaction. This will make payment process smoother and faster and increase convenience for customers, minimize financial reconciliation errors, and cut down checkout time. This will also make the whole process more streamlined and minimize the risk of revenue leakage in the financial space.

- **Customer Satisfaction**

Customer satisfaction is very important for the success of any business firm. Customer satisfaction scores today average 75%, which complains long wait time, inconsistent order delivery and fragmented payment process. The SAP SD module implementation is expected to increase the customer satisfaction scores by 20% to 90% by the end of first year. This increased customer retention will result in more people coming back, more people referring people, helping to make the food court a better place and improving the reputations of its shops and the food court.

- **Resource Utilization (Food Waste Reduction)**

Inefficient resource use needs to be optimized, in which case the greatest impact should be in the reduction of food waste. Currently, 15% of food is wasted in the food court on average because there are still inefficient inventory management and no real-time demand forecasting. Using the real time tracking of inventory and demand forecasting capabilities of the SAP SD module will reduce food waste by 15% to 12.75%. This will provide cost savings, increased profitability, and sustainability of the operation. Additionally, the food court's environmental sustainability goals are also addressed by SDG 12 (Responsible Consumption and Production).

- **Operational Costs**

At the same time, reducing operational costs is a priority. Inefficient in order and payment processing and resource wastage processes result in unnecessarily high operational costs. To reduce operational costs by 10% the SAP SD module generates expected decreased in workflow, automation of repetitive tasks, and improved utilization of resources. After reducing costs, financial sustainability will increase to allow the reinvestment in customer experience improvement and in future technological advancements.

- **Staff Productivity**

Finally, staff productivity will increase tremendously. At present, hand entry of orders, payment processing, and disjointed workflow slows staff productivity further and generates the stress of extra work. It is expected to increase staff productivity by 15% through automation of repetitive tasks as well as better workflow coordination to free your employees' time to focus on higher value tasks such as customer service and food preparation. This will encourage a more motivated and efficient workforce that will produce more positive effects on the overall quality of service and create a good work environment.

7.0 Conclusion

The implementation of SAP ERP system at Madam Yan Food Court will be the solution to solve the existing operational challenges. SAP Sales and Distribution (SD) module provides a comprehensive solution by centralizing order processing, integrating payment systems and streamlining workflows which help to enhance both operational efficiency and customer satisfaction. The proposed ERP solution simplifies ordering and payment process. Other than

that, it also ensures a better coordination between food and drinks preparation. The solution resolves the problem while positioning Madam Yan Food Court for a sustainable growth and customer satisfaction.

The benefit of the proposed solution is the order processing time reduced and the payment efficiency improved which will optimize resource utilization. Moreover, the SAP SD module corresponds with various United Nations Sustainable Development Goals (SDGs), highlighting sustainability, social responsibility, and innovation. The food court may improve sustainability and equity by reducing food waste, boosting worker productivity, and promoting inclusive payment systems.

The phased implementation plan ensures a smooth transition to the new system, providing adequate time for testing, personnel training, and continuous improvement. Madam Yan Food Court may enhance its operations and customer experience by leveraging data-driven insights from the SAP SD module. The deployment of the SAP SD module will position Madam Yan Food Court as a modern, efficient, and customer-focused enterprise. It will become a standard for other food courts in the food sector. As the food court industry moves forward with this implementation, the SAP SD module will address challenges and provide scalability for future growth and adaptation to evolving customer needs. To ensure the long-term success of the system, regular audits and key performance indicators (KPIs) should be implemented to monitor its effectiveness, allowing for data-driven optimizations. Additionally, continuous IT support and system maintenance should be prioritized, including user training and timely updates, to maximize operational efficiency and address any technical challenges.

References

Vaka, D. K. (2023). Empowering Food and Beverage Businesses with S/4HANA: Addressing Challenges Effectively. *Journal of Artificial Intelligence, Machine Learning and Data Science*, 1(1), 376–381. <https://doi.org/10.51219/jaimld/dilip-kumar-vaka/96>

Schwarz, L. (2024, December 6). *6 Key phases of an ERP Implementation Plan*. Oracle NetSuite. <https://www.netsuite.com/portal/resource/articles/erp/erp-implementation-phases.shtml>

Rodina, D. (n.d.). *Burger Restaurant (BPMN Diagram) - Software Ideas Modeler*. <https://www.softwareideas.net/a/1607/burger-restaurant-bpmn-diagram->

Appendix

The clear image for BPMN business process diagram

